

Research Assignment: Investigating the Role, Impact, and Relevance of UML in Software Engineering in Modern Software Engineering

1 Why UML Exists

1. **Why was UML created in Software Engineering**
 - What problems was it trying to solve?
 - Were developers struggling before UML?
2. **Could software engineering function effectively without UML**
Explain your reason.
3. **Why do engineers model systems before implementation?**
Is coding immediately always a good approach? Explain!

2 Impact of UML

1. **How does UML improve communication among stakeholders?** (Developer, Manager, Client, Tester)
2. **How Does UML Reduce Software Development Risks?**
Example:
 - Misunderstood requirements
 - Wrong architecture
 - Duplicated modules
 - Late project delivery
 - Costly redesign
3. **How does UML improve the maintainability of software**
Imagine another developer joins after 2 years. How does UML help?
4. **How does UML support scalability?**
Can a system grow more easily when properly modeled?

3 If UML Does Not Exist

1. **What would happen if software projects were developed without UML or any modeling approach?**
Possible angles:

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- Communication breakdown
 - Requirement confusion
 - Messy system structure
 - Integration failure..etc.

2. Which Projects can survive without UML? Why?

- Small apps?
- Startup MVP?
- Enterprise banking system?
- Hospital System?

4 Reflection

Do you believe UML is still important in Agile/AI-assisted modern software development? why or why not?